

LEDGER · 2026-09-04 · AGRICULTURAL CHEMISTRY

URCASU Mechanochemical Fertilizer via Twin-Screw Extrusion

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Polymer-Coated Urea (Turf-Maker Duration 44-0-0)
Entry application	Professional Golf Course and Sports Turfgrass Nitrogen Management
Measured advantage	Claimed 0% microplastic residue (categorical) and 'significantly higher NDVI', but lacks a quantified 2x step-change performance delta over polymer-coated urea.
Time to first revenue	4 months
Gross margin at parity	74.9%
Startup CapEx	\$110,500
Payback from first sale	2.3 months

Capital productivity

9.9x

Regulatory risk

Low — no material regulatory CapEx reported (\$0 estimated)

Market absence

Verified — the search is reported in the dossier: channels searched, companies searched, closest substitutes named

Why it failed

- Voided by: Negative control / sub-2x
- Evidence rests on non-peer-reviewed sources

Assessment

The concept demonstrates strong scientific backing, excellent unit economics, low CapEx (\$110,500), and rapid time-to-first-revenue. However, it fails the superiority criterion because microplastic elimination is an environmental preference attribute, and the agronomic performance claim lacks a quantified 2x step-change multiplier against polymer-coated urea.

Also flagged

- Duplicate references inflating the evidence base
- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-04 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

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